

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 8 — Energetics

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. A chemical reaction that proceeds with the evolution of heat is called: ✓ **B — Exothermic reaction**

1 mark

An **exothermic reaction** releases heat to the surroundings; the system loses energy so ΔH is negative. Endothermic reactions absorb heat (positive ΔH). Thermochemical reaction is the broader category, not the specific type that releases heat.

Q2. If the ΔH value of a reaction is negative, the reaction is: ✓ **B — Exothermic** 1 mark

By the sign convention in Section 8.3: ΔH is **negative for exothermic** reactions (heat flows out of the system) and positive for endothermic reactions. A common student error is to confuse the sign, believing negative ΔH means the reaction absorbs energy.

Q3. $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$, $\Delta H^\circ = -393.5 \text{ kJ}$. This reaction is: ✓ **B — Exothermic** 1 mark

The negative sign on $\Delta H^\circ = -393.5 \text{ kJ}$ confirms energy is released to the surroundings. This is also the textbook example given in Section 8.2.1 for combustion of carbon, a classic exothermic reaction.

Q4. Bond breaking is always: ✓ **B — Endothermic** 1 mark

Section 8.4 states: "Breaking bonds always **requires energy**" (endothermic), while forming bonds always releases energy (exothermic). Students often get this backwards, thinking breaking bonds releases energy.

Q5. The enthalpy change of a reaction equals: ✓ **C — Sum of BDE of reactants minus sum of BFE of products** 1 mark

The textbook formula (Section 8.4): $\Delta H^\circ = \text{Sum of bond dissociation energies of reactants} - \text{Sum of bond formation energies of products}$. Option D reverses the subtraction, a common sign error that students make.

Q6. Activation energy must be _____ the average kinetic energy: ✓ **C — greater than** 1 mark

Section 8.5 (exercise Q(v)) states activation energy must be **greater than** the average kinetic energy of the reacting molecules for an effective collision to occur. If particles do not have enough energy, collisions are ineffective and no reaction takes place.

Q7. A catalyst increases the rate of a reaction by: ✓ **C — lowering activation energy** 1 mark

Section 8.6: a catalyst provides a new reaction mechanism with **lower activation energy**. It does not change the overall enthalpy (ΔH remains the same). Students often wrongly choose "increasing enthalpy," confusing the energy diagram peak with the overall energy change.

Q8. A balanced chemical equation that also shows the heat change is called: ✓ **B — Thermochemical equation** 1 mark

Section 8.2 definition: "A balanced chemical equation which also shows heat change of a chemical reaction is called **thermochemical equation**." The branch of chemistry studying these is thermochemistry.

Q9. Which product is NOT released in aerobic respiration? ✓ **C — Lactic acid** 1 mark

Lactic acid is the product of *anaerobic* respiration ($\text{Glucose} \rightarrow \text{Lactic acid} + \text{Energy}$). Aerobic respiration produces carbon dioxide, water, and energy. Lactic acid builds up in muscles during intense exercise when oxygen supply is insufficient.

Q10. Which reaction has a positive enthalpy change? ✓ **C — $\text{N}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{NO}_{(g)}$, $\Delta H^\circ = +180.5 \text{ kJ}$**

1 mark

A **positive ΔH** indicates an endothermic reaction. Options A, B, and D all have negative ΔH values (exothermic). The formation of NO from N_2 and O_2 absorbs energy (+180.5 kJ), making it endothermic, as listed in Section 8.2.2 of the textbook.

Q11. Lipids act as reserve energy stores because: ✓ **B — they can store very large amounts of energy**

1 mark

Section 8.7 states: "Lipids can store very **large amounts of energy** in our body." When glucose oxidation is insufficient during intense exercise, the body oxidizes lipids for energy. Lipids are not catalysts or water-soluble compounds.

Q12. In living organisms, the catalysts that catalyze chemical reactions are called: ✓ **C — Enzymes**

1 mark

Section 8.6 states: "In the bodies of living organisms **enzymes** catalyze chemical reactions." Enzymes are biological catalysts that lower activation energy for biochemical reactions. Vitamins and hormones serve different roles; lipids are energy stores.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) — Chemical energetics and thermochemical reactions **3 marks**

Chemical energetics: The study of energy changes in chemical reactions is called **chemical energetics**.

1 mark

Thermochemical reaction: A chemical reaction which proceeds with the evolution or absorption of heat is called a **thermochemical reaction**. A balanced chemical equation that also shows the heat change is called a **thermochemical equation**.

1 mark

Example: $\text{C}_{(s)} + \text{O}_{2(g)} \rightarrow \text{CO}_{2(g)}$ $\Delta H^\circ = -393.5 \text{ kJ}$

1 mark

OR

System: The part of the universe that we want to focus our attention on is called the **system**. In chemistry, the system is usually the substance that changes physically or chemically.

1 mark

Surroundings: The rest of the universe (everything around the system) is called the **environment** or surroundings.

1 mark

Example: When studying the reaction of limestone and hydrochloric acid in a test tube, the limestone and HCl solution form the **system**. The test tube, heat source, and everything around it form the **surroundings**.

1 mark

2(ii) — Exothermic and endothermic reactions with examples **3 marks**

Exothermic reaction: A chemical reaction that proceeds with the **evolution of heat** is called an exothermic reaction. The chemical system transfers energy to the surroundings; ΔH is negative.

1 mark

Examples:

2 marks

- $\text{C}_{(s)} + \text{O}_{2(g)} \rightarrow \text{CO}_{2(g)}$ $\Delta H^\circ = -393.5 \text{ kJ}$
- $2\text{H}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{H}_2\text{O}_{(g)}$ $\Delta H^\circ = -571.6 \text{ kJ}$

OR

Endothermic reaction: A chemical reaction that proceeds with the **absorption of heat** is called an endothermic reaction. Heat is transferred from surroundings to the system; ΔH is positive.

1 mark

Examples:

2 marks

- $\text{H}_{2(g)} + \text{I}_{2(g)} \rightarrow 2\text{HI}_{(g)}$ $\Delta H^\circ = +53.8 \text{ kJ}$
- $\text{N}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{NO}_{(g)}$ $\Delta H^\circ = +180.5 \text{ kJ}$

2(iii) — Sign convention for enthalpy change 3 marks

The amount of heat or thermal energy evolved or absorbed in a chemical reaction is called **enthalpy of reaction**. 1 mark

Sign convention:

- **ΔH is negative** for exothermic reactions: the system releases energy to the surroundings, so the system loses heat (energy decreases). 1 mark
- **ΔH is positive** for endothermic reactions: the system absorbs energy from the surroundings, so the system gains heat (energy increases). 1 mark

OR

Standard enthalpy change (ΔH°): Enthalpy of reaction measured at **25°C (298 K)** and **one atmospheric pressure** is known as standard enthalpy change. 1 mark

It is denoted by the symbol **ΔH°** . 1 mark

Example: $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$, $\Delta H^\circ = -393.5 \text{ kJ}$ 1 mark

2(iv) — Classify processes as exothermic or endothermic 3 marks

- (a) **Freezing of water:** Exothermic (heat is released when liquid water converts to solid ice). 1 mark
- (b) **Combustion of methane:** Exothermic (burning releases large amounts of heat energy). 1 mark
- (c) **Decomposition of limestone** ($CaCO_3 \rightarrow CaO + CO_2$): Endothermic (heat must be supplied to break down limestone). 1 mark

OR

- (a) **Sublimation of dry ice:** Endothermic (solid CO_2 absorbs heat from surroundings to convert directly to gas). 1 mark
- (b) $H_2O_{(g)} \rightarrow H_2O_{(l)}$: Exothermic (condensation releases heat, as steam gives out energy when it becomes liquid). 1 mark
- (c) **Burning of fuels:** Exothermic (combustion releases heat and light energy to the surroundings). 1 mark

2(v) — Bond dissociation energy and bond formation energy 3 marks

Bond dissociation energy (BDE): The amount of energy required to break one mole of a particular bond to form neutral atoms is called the **bond dissociation energy**. Bond breaking always *absorbs* energy (endothermic). 1 mark

Bond formation energy (BFE): The amount of energy released when neutral atoms form one mole of a bond is called the **bond formation energy**. Bond forming always *releases* energy (exothermic). 1 mark

Relationship: $\Delta H^\circ = \text{Sum of BDE of reactants} - \text{Sum of BFE of products}$. The *difference* between these two determines whether the overall reaction absorbs or releases energy. 1 mark

OR

Bond breaking absorbs energy because energy must be put in to overcome the attraction between atoms. This is an endothermic process. 1 mark

Bond forming releases energy because when atoms form a bond, they move to a lower energy state, releasing the energy difference. This is an exothermic process. 1 mark

Overall: If the energy absorbed in breaking bonds (reactants) is *less than* the energy released in forming bonds (products), the net reaction is exothermic ($\Delta H < 0$). If breaking bonds requires *more* energy than is released in forming bonds, the reaction is endothermic ($\Delta H > 0$). 1 mark

2(vi) — ΔH formula and bond energy calculation 3 marks

Formula: $\Delta H^\circ = \text{Sum of bond dissociation energies of reactants} - \text{Sum of bond formation energies of products}$ 1 mark

(a) If sum of BDE of reactants > sum of BFE of products: **ΔH is positive** (endothermic reaction, net energy absorbed). 1 mark

(b) If sum of BDE of reactants < sum of BFE of products: **ΔH is negative** (exothermic reaction, net energy released). 1 mark

OR

Calculation ($\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$): 3 marks

Equation: $\text{H}-\text{H} + \text{I}-\text{I} \rightarrow 2 \text{H}-\text{I}$
 $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$
 $\Delta H = [\text{BDE}(\text{H}-\text{H}) + \text{BDE}(\text{I}-\text{I})] - [2 \times \text{BFE}(\text{H}-\text{I})] = [436 + 151] - [2 \times 299] = 587 - 598 = -11 \text{ kJ/mol}$
Result: $\Delta H = -11 \text{ kJ/mol}$ (Exothermic, negative value)

Bonds broken (reactants): $\text{H}-\text{H}$ (436) + $\text{I}-\text{I}$ (151) = 587 kJ. Bonds formed (products): $2 \times \text{H}-\text{I}$ (299) = 598 kJ. Since $598 > 587$, more energy is released than absorbed, so ΔH is negative and the reaction is exothermic.

2(vii) — Activation energy and collision theory 3 marks

Activation energy: The minimum amount of energy that, in addition to the average kinetic energy, particles must have in effective collisions is called the **activation energy**. 1 mark

Effective collision: A collision in which reacting particles have sufficient energy (at least equal to activation energy) AND correct orientation, causing bonds to break and new bonds to form. 1 mark

Ineffective collision: A collision in which particles do not have enough energy or do not have the correct orientation, so the particles simply bounce off without reacting. 1 mark

OR

Two conditions for an effective collision: (1) The colliding particles must have energy **equal to or greater than the activation energy**. (2) The particles must collide with the **correct orientation** so that the atoms needing to form new bonds actually collide with each other. 2 marks

Higher activation energy and slower rate: If activation energy is higher, fewer particles in a sample will have enough energy to react at any given moment, so fewer effective collisions occur per second, resulting in a slower reaction rate. 1 mark

2(viii) — Catalyst and catalysis 3 marks

Catalyst: A substance that accelerates a chemical reaction but remains **chemically unchanged** at the end of the reaction is called a catalyst. 1 mark

Catalysis: The phenomenon in which the rate of a chemical reaction is increased by a catalyst is called catalysis. 1 mark

Mechanism: The catalyst provides a new reaction pathway with **lower activation energy**. More particles can now reach the activation energy threshold, so more effective collisions occur per second. Importantly, the catalyst does *not* affect the overall thermodynamics or enthalpy of the reaction; the energy difference between reactants and products (i.e., ΔH) stays the same. 1 mark

OR

Enzyme: In the bodies of living organisms, the biological catalysts that catalyze chemical reactions are called **enzymes**. 1 mark

Difference from industrial catalysts: Enzymes are highly **specific**, meaning each enzyme typically catalyzes only one type of reaction or works on only one type of substrate molecule. Industrial chemical catalysts (e.g., iron in the Haber process) are generally less specific and can catalyze a range of reactions under appropriate conditions. 2 marks

2(ix) — Respiration as an exothermic process 3 marks

Respiration: Respiration is a biochemical process in which energy is released from food in a biological system. During this process, glucose is oxidized in the body of living organisms and energy is released.

1 mark

Why exothermic: In respiration, glucose is **oxidized** (bonds broken and new bonds formed), and the energy released in forming products (CO_2 , H_2O) is greater than the energy absorbed in breaking the bonds of reactants (glucose, O_2). The net result is a release of energy, so ΔH is negative, making it **exothermic**.

2 marks

OR

Aerobic respiration word equation:

Glucose + Oxygen → Carbon dioxide + Water + Energy 1 mark

Reactants: Glucose (food / stored chemical energy) and Oxygen (from breathing). 1 mark

Products: Carbon dioxide (exhaled), Water (released), and Energy (used for all body functions). The energy comes from the breaking of chemical bonds in glucose, with more energy released in forming CO_2 and H_2O bonds than was absorbed. 1 mark

2(x) — Aerobic vs anaerobic respiration 3 marks

Aerobic respiration: Respiration that **requires oxygen** to break down glucose to release energy. 1 mark

Word equation: Glucose + Oxygen → Carbon dioxide + Water + Energy

Anaerobic respiration: Respiration that **does not require oxygen** to break down glucose to release energy. 1 mark

Word equation: Glucose → Lactic acid + Energy 1 mark

OR

Aerobic releases more energy than anaerobic respiration because complete oxidation of glucose in the presence of oxygen releases more energy than the partial breakdown that occurs without oxygen. 1 mark

Role of lipids during intense exercise: During intense exercise, the body's demand for energy is very high and the oxidation of glucose alone may not be sufficient. At this point, **lipids are oxidized for energy**. Lipids act as **reserve energy stores** because they can store very large amounts of energy in the body, providing a secondary fuel source when glucose is depleted. 2 marks

2(xi) — Reaction pathway diagram features 3 marks

For an **exothermic reaction** pathway diagram:

- (a) **Reactants level:** starting energy level on the left, at a higher position than products. 1 mark
- (b) **Products level:** ending energy level on the right, at a *lower* position than reactants (energy released). 1 mark
- (c) **Activation energy:** the energy from the reactants level up to the peak (transition state) at the top of the curve. half
- (d) **Enthalpy change (ΔH):** the difference in energy between reactants and products, shown as a downward arrow on the right side of the curve; ΔH is **negative**. half

OR

For an **endothermic reaction** pathway diagram:

- (a) **Reactants level:** starting energy level on the left, at a *lower* position than products.
- (b) **Products level:** ending energy level on the right, *higher* than reactants (energy absorbed).
- (c) **Activation energy:** energy from reactants level up to the transition state peak.
- (d) **Enthalpy change (ΔH):** difference in energy between products and reactants, shown as an upward arrow; ΔH is **positive** because products have more energy than reactants.

3 marks for all four correctly labeled with sign of ΔH stated

3(i) — Exothermic/endothermic reactions with thermochemical equations 5 marks

(a) Definitions and sign convention [2 marks]

Exothermic reaction: A chemical reaction that proceeds with the evolution of heat. The system transfers energy to the surroundings; **ΔH is negative.** 1 mark

Endothermic reaction: A chemical reaction that proceeds with the absorption of heat. Heat is transferred from surroundings to the system; **ΔH is positive.** 1 mark

(b) Thermochemical equations [3 marks]

Exothermic (1 example): $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$, $\Delta H^\circ = -393.5 \text{ kJ}$ (negative ΔH : heat evolved) 1 mark

Endothermic (2 examples):

• $H_{2(g)} + I_{2(g)} \rightarrow 2HI_{(g)}$, $\Delta H^\circ = +53.8 \text{ kJ}$ (positive ΔH : heat absorbed) 1 mark

• $N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$, $\Delta H^\circ = +180.5 \text{ kJ}$ (positive ΔH : heat absorbed) 1 mark

OR

(a) Thermochemical reaction and thermochemistry [2 marks]

Thermochemical reaction: A chemical reaction which proceeds with the evolution or absorption of heat. A balanced chemical equation that also shows the heat change is called a **thermochemical equation.**

1 mark

Thermochemistry: The branch of chemistry which deals with heat or thermal energy changes in chemical reactions. Textbook example: $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$, $\Delta H^\circ = -393.5 \text{ kJ}$. 1 mark

(b) Classification of five reactions [3 marks]

- (i) $C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)}$, $\Delta H^\circ = -110.5 \text{ kJ}$: **Exothermic** (negative ΔH)
- (ii) $N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$, $\Delta H^\circ = +180.5 \text{ kJ}$: **Endothermic** (positive ΔH)
- (iii) Freezing of water: **Exothermic** (heat released as liquid becomes solid)
- (iv) Sublimation of dry ice: **Endothermic** (heat absorbed from surroundings)
- (v) Combustion of methane: **Exothermic** (burning releases heat)

1 mark per correct pair (type + justification), up to 3 marks

3(ii) — Bond energy calculation: $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ 5 marks

(a) Formula and explanation [2 marks]

$\Delta H^\circ = \text{Sum of BDE of reactants} - \text{Sum of BFE of products}$ 1 mark

BDE (Bond Dissociation Energy): Energy absorbed to break one mole of a bond into neutral atoms (endothermic, energy input). **BFE (Bond Formation Energy):** Energy released when neutral atoms form one mole of a bond (exothermic, energy released). The difference determines whether the overall reaction is exo- or endothermic. 1 mark

(b) Calculation [3 marks]

Balanced equation: $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$ Bond data: $\text{H-H} = 436 \text{ kJ/mol}$ $\text{Cl-Cl} = 243 \text{ kJ/mol}$ $\text{H-Cl} = 432 \text{ kJ/mol}$ Bonds BROKEN (reactants): $1 \times \text{H-H} = 1 \times 436 = 436 \text{ kJ}$ $1 \times \text{Cl-Cl} = 1 \times 243 = 243 \text{ kJ}$ Total BDE = $436 + 243 = 679 \text{ kJ}$ [1 mark] Bonds FORMED (products): $2 \times \text{H-Cl} = 2 \times 432 = 864 \text{ kJ}$ [1 mark] $\Delta H = 679 - 864 = -185 \text{ kJ/mol}$ [1 mark] Result: $\Delta H = -185 \text{ kJ/mol} \Rightarrow \text{EXOTHERMIC}$ (More energy released in forming HCl bonds than absorbed in breaking H-H and Cl-Cl bonds)

OR

(a) BDE and BFE definitions [2 marks]

Bond dissociation energy: The amount of energy required to break one mole of a particular bond to form neutral atoms. Bond breaking **absorbs energy** (endothermic). 1 mark

Bond formation energy: The amount of energy released when neutral atoms form one mole of a bond. Bond forming **releases energy** (exothermic). 1 mark

(b) Calculation: $\text{H}_2 + \text{F}_2 \rightarrow 2\text{HF}$ [3 marks]

Balanced equation: $\text{H}_2(\text{g}) + \text{F}_2(\text{g}) \rightarrow 2\text{HF}(\text{g})$ Bond data: $\text{H-H} = 436 \text{ kJ/mol}$ $\text{F-F} = 155 \text{ kJ/mol}$ $\text{H-F} = 567 \text{ kJ/mol}$ Bonds BROKEN (reactants): $1 \times \text{H-H} = 436 \text{ kJ}$ $1 \times \text{F-F} = 155 \text{ kJ}$ Total BDE = $436 + 155 = 591 \text{ kJ}$ [1 mark] Bonds FORMED (products): $2 \times \text{H-F} = 2 \times 567 = 1134 \text{ kJ}$ [1 mark] $\Delta H = 591 - 1134 = -543 \text{ kJ/mol}$ [1 mark] Result: $\Delta H = -543 \text{ kJ/mol} \Rightarrow \text{EXOTHERMIC}$ (Large energy release due to very strong H-F bonds)

3(iii) — Activation energy, collision theory, and reaction pathway diagram 5 marks

(a) Activation energy and collision theory [3 marks]

Activation energy: The minimum amount of energy that, in addition to the average kinetic energy, particles must have in effective collisions is called the **activation energy**. No reaction occurs if the energy of the reacting particles is lower than the activation energy. 1 mark

Effective collision: A collision is effective only when (i) particles possess energy at least equal to the activation energy AND (ii) the particles have correct orientation so that the atoms needing to form new bonds collide with each other directly. This produces an unstable activated complex at the transition state, which then breaks down to give products. 1 mark

Ineffective collision: If particles lack sufficient energy or have wrong orientation, they simply bounce off one another without reacting. The higher the activation energy, the fewer particles have enough energy and the slower the reaction rate. 1 mark

(b) Reaction pathway diagram for exothermic reaction [2 marks]

Energy | Transition state (peak) | / \ | / \ <-- Ea without catalyst (higher) | / Ea(unc) \ |-----/
 \ | Reactants \ <-- Ea with catalyst (lower peak) | \----- | DeltaH(-) Products (lower than
 reactants) | +-----> Reaction progress Labels required:
 Reactants, Products, Transition state, Activation energy (uncatalyzed), Activation energy
 (catalyzed), Enthalpy change (DeltaH, negative, downward arrow)

1 mark for correct relative positions (reactants higher than products, peak above both). 1 mark for all six labels correctly placed.

OR

(a) Catalyst definition and mechanism [3 marks]

Catalyst: A substance that accelerates a chemical reaction but remains **chemically unchanged** at the end of the reaction. The phenomenon is called catalysis. 1 mark

How it increases rate without changing ΔH : The catalyst provides a new reaction pathway with **lower activation energy**. Because the energy barrier is lower, more molecules in the sample have sufficient energy to react, so more effective collisions occur per unit time and the rate increases. However, the reactants and products remain the same and their energy levels are unchanged, so the overall **enthalpy change (ΔH) is not affected**. 2 marks

(b) Reaction pathway diagram for endothermic reaction [2 marks]

Energy | Transition state (peak) | / \ | / \ Ea \ | / \----- |-----/ Reactants Products (higher
 than reactants) | | DeltaH = positive (upward arrow from reactants to products) | +-----
 -----> Reaction progress Labels: Reactants (lower), Products (higher), Transition
 state, Activation energy (reactants to peak), DeltaH (positive, upward)

Sign of ΔH : Positive, because products are at a higher energy level than reactants. Energy has been absorbed from the surroundings. 2 marks for diagram + sign explanation

3(iv) — Respiration: exothermic process, types, lipids 5 marks

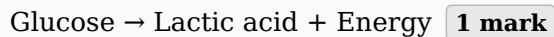
(a) Respiration, exothermic nature, aerobic vs anaerobic 3 marks

Respiration: Respiration is a biochemical process in which energy is released from food (glucose) in a biological system. Glucose is oxidized and energy is released; therefore, respiration is an **exothermic process** (ΔH is negative). 1 mark

Aerobic respiration requires oxygen to break down glucose:



Anaerobic respiration does not require oxygen:



(b) Role of lipids as reserve energy stores 2 marks

Lipids are important substances for building our body. They also act as **reserve energy sources** because lipids can store very large amounts of energy in our body. When a person exercises intensely, the oxidation of glucose is not enough to supply all the energy required. At this stage, **lipids are oxidized for energy**, providing an additional fuel source to maintain body functions during prolonged or intense activity.

2 marks

OR

(a) Comparison of aerobic and anaerobic respiration 3 marks

Heading	Aerobic Respiration	Anaerobic Respiration
Oxygen requirement	Requires oxygen	Does not require oxygen
Reactants	Glucose + Oxygen	Glucose only
Products	Carbon dioxide + Water + Energy	Lactic acid + Energy
Energy released	More energy released	Less energy released

3 marks (1 per correct row, up to 3)

(b) Enzymes as biological catalysts and activation energy 2 marks

Enzymes are biological molecules (proteins) that act as **catalysts** in living organisms. Like any catalyst, enzymes lower the **activation energy** of biochemical reactions, allowing them to proceed rapidly at body temperature (37°C) without requiring the high temperatures that industrial reactions often need. Each enzyme is highly specific, catalyzing only one reaction or type of reaction. Without enzymes, the activation energy for most biochemical reactions would be too high for them to occur at the speeds necessary to sustain life. 2 marks